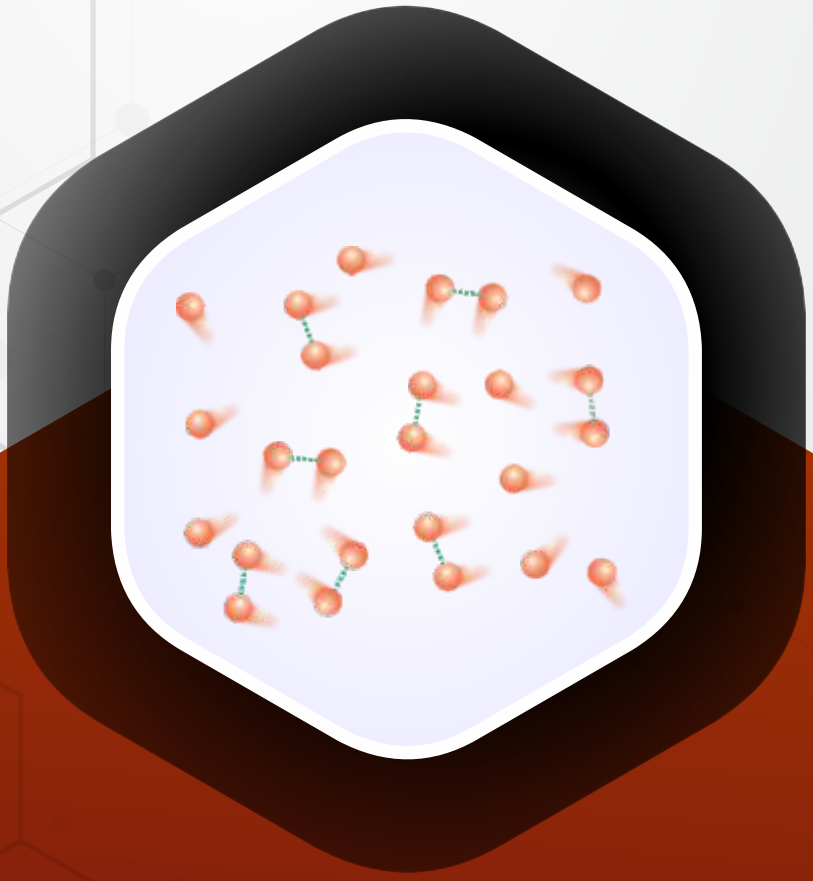
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Real Gas

(English Medium)

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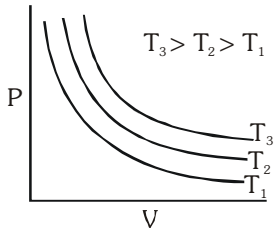
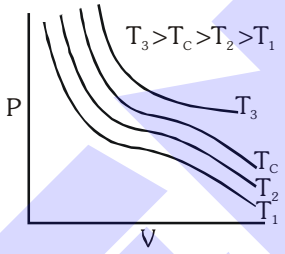
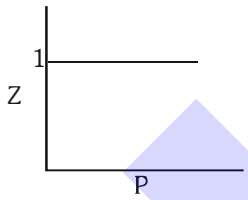
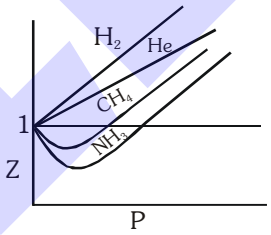
REAL GAS

1. Introduction :

An ideal gas is a hypothetical gas whose pressure, volume and temperature behaviour is completely described by the ideal gas equation. Actually no gas is ideal or perfect in nature. All gases are real gases.

Real gases do not obey the ideal gas laws exactly under all conditions of temperature and pressure.

1.1 COMPARISION BETWEEN REAL AND IDEAL GAS :

	IDEAL GAS	REAL GAS
(i)	$PV = nRT$	$PV \neq nRT$
(ii)	No liquifaction is possible.	Liquifaction is possible below a certain temperature.
		
		⇒ Follow critical phenomena and can not liquefy above T_C.
(iii)		
(iv)	No interaction force is present between gas particles.	Interaction force exist between gas particles which vary, depending upon conditions.
(v)	Volume of gas particles is negligible w.r.t. volume of container.	Volume of gas particles has significant value and can not be neglected normally w.r.t. volume of container.

1.2 REASON OF DEVIATION IN GASES FROM IDEAL BEHAVIOUR :

Real gases deviates from ideal behaviour because of mainly two assumptions of "Kinetic theory of gases".

(i) The volume of gas particle is negligible compared to volume of container (while the real gas particle may have some significant volume).

(ii) There is no interaction between gaseous particles (while intermolecular forces exist between real gas particles).

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1.3 COMPRESSIBILITY FACTOR (Z) :

The extent to which a real gas departs from the ideal behaviour may be expressed in terms of compressibility factor or compression factor (Z),

$$Z = \frac{(PV)_{real}}{(PV)_{ideal}} = \left(\frac{V_{m,\,real}}{V_{m,\,ideal}}\right)_P = \frac{PV_m}{RT} \qquad [V_m = \text{molar volume}]$$

1.3.1 Compressibility factor vs pressure :

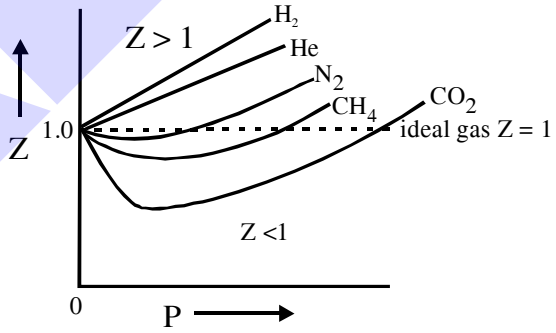
(i) For an ideal gas, Z = 1 and is independent of temperature and pressure.

(ii) Exceptional behaviour of H_2 and He :
For these gases Z > 1 at 0ºC.

(iii) Effect of pressure :
(a) At very low P, $PV_m \approx RT$ i.e. $Z \approx 1$
(b) At low P, $PV_m < RT$ i.e. $Z < 1$
$\Rightarrow$ attractive forces dominant
$\Rightarrow$ Negative deviation from ideal behaviour.
$\Rightarrow$ Gas is more compressible than ideal behaviour.
(c) At high P, $PV_m > RT$ i.e. $Z > 1$
$\Rightarrow$ repulsive forces dominant
$\Rightarrow$ Positive deviation from ideal behaviour.
$\Rightarrow$ Gas is less compressible than ideal behaviour.

(iv) For the gases which are easily liquefied (e.g. CO_2) Z dips sharply below the ideal line in the low pressure region.

(v) Effect of temperature : An increase in temperature shows a decrease in deviation from ideal behaviours, Z approaches unity with increase in temperature.



1.4 VAN DER WAAL EQUATION OF REAL GASES :

The ideal gas equation does not consider the effect of attractive forces and molecular volume.

Van der Waal corrected the ideal gas equation by taking the effect of

(a) Molecular volume

(b) Molecular attraction

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(A) Volume correction :

In the ideal gas equation, $P_iV_i = nRT$, V_i represents the ideal volume where the molecules can move freely. In real gases, a part of the total volume is occupied by the gas molecules. Hence the free volume V_i is the total volume V minus the volume occupied by the gas molecules.

Real volume of gas = Actual volume of container – volume occupied by molecules in motion.

$\boxed{V_i = V - nb}$ for n mole of gas

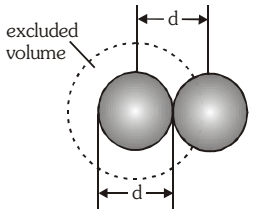


Illustration of excluded volume

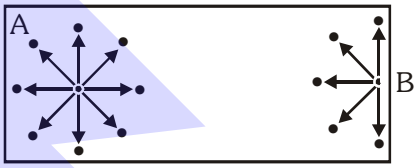
Where b is termed the 'excluded volume' or 'co-volume' per mole.
It is constant and characteristic for each gas.

$\boxed{b = 4 \times \text{volumes of one molecules} \times N_A}$

(B) Pressure correction :

In order to take account the effect of intermolecular forces of attraction, let us consider a molecule A in the midst of the vessel.

This molecule is surrounded by other molecules in a symmetrical manner and is being attracted uniformly on all sides by the neighbouring molecules with the result that this molecule on the whole experiences no net force of attraction.



Now, consider a molecule B near the side of the vessel, which is about to strike one of its sides, thus contributing towards the total pressure of the gas. There are molecules only in one side of the vessel, i.e. towards its centre, with the result of that, this molecule experiences a net force of attraction towards the centre of the vessel. This results in decreasing the velocity of the molecule, and hence its momentum. Thus, the molecule does not contribute as much force as it would have, had there been no force of attraction. Thus, the pressure of a real gas would be smaller than the corresponding pressure of an ideal gas.

Van der Waals noted that the total force of attraction on any molecule about to hit a wall is proportional to the concentration of neighbouring molecules, n/V. However, the number of molecules about to hit the wall per unit wall area is also proportional to the concentration n/V. Therefore, the force per unit wall area, or pressure, is reduced from that assumed in the ideal gas wall by a factor proportional to n^2/V^2. Letting a be the proportionality constant, we can write

$$P\ (\text{actual}) = P(\text{ideal}) - \frac{an^2}{V^2}$$

or $$P\ (\text{ideal}) = P(\text{actual}) + \frac{an^2}{V^2}$$

or $$P_i = P + \frac{an^2}{V^2}$$

'a' is a constant which depends upon the nature of the gas, Greater the intermolecular attraction, greater the volume of 'a'.

Combining the two corrections,

$\Rightarrow \quad P_i \cdot V_i = nRT$

or, $$\boxed{\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT}$$

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For 1 mole of gas

$$\left(P+\frac{a}{V^2}\right)(V-b)=RT$$

1.4.1 The constants 'a' and 'b' :

Van der Waals constant for attraction 'a' and volume 'b' are characteristic constants for a given gas, independent from pressure and temperature conditions.

(i) The 'a' values for a given gas are measure of intermolecular forces of attraction. More are the intermolecular forces of attraction, more will be the value of a.

(ii) The gas having higher value of 'a' can be liquefied easily.

(iii) Unit of 'a' is atm L^2 $mole^{-2}$ or dyne cm^4 $mole^{-2}$ or Nm^4mol^{-2}

(iv) Unit of 'b' is L $mole^{-1}$ or cm^3 $mole^{-1}$ or m^3 mol^{-1}

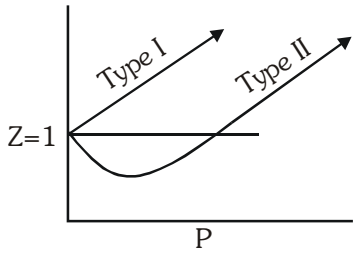
1.4.2 The van der waal's constants for some common gases

Gas	a ($atmL^2mol^{-2}$)	b ($L\ mol^{-1}$)
Ammonia	4.17	0.0371
Argon	1.337	0.0322
Carbon dioxide	3.59	0.0427
Carbon monoxide	1.49	0.0399
Chlorine	6.49	0.0562
Ethane	5.49	0.0638
Ethanol	2.56	0.087
Ethylene	4.552	0.0582
Helium	0.034	0.0237
Hydrogen	0.024	0.0266
Hydrogen chloride	3.67	0.0408
Hydrogen bromide	4.45	0.0433
Methane	2.25	0.0428
Neon	0.21	0.0171
Nitric oxide	1.34	0.0279
Nitrogen	1.39	0.0319
Oxygen	1.36	0.0318
Sulphur dioxide	3.71	0.0564
Water	5.44	0.0305

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1.4.3 Verification of compressibility factor using Van der Waal's equation :

Variation of Z with P for real gas at any temperature is given by following graph.



Van der waal equation :

$$\left(P+\frac{a}{V_m^2}\right)(V_m-b)=RT \qquad \text{..... (1)}$$

(i) At low pressure and constant temperature

At low pressure, V_m will be high hence 'b' can be neglected in comparision to V_m. But $\frac{a}{V_m^2}$ can't be neglected as pressure is low. Thus equation (1) reduces to

$$\left(P+\frac{a}{V_m^2}\right)V_m=RT$$

$$PV_m+\frac{a}{V_m}=RT$$

$$\frac{PV_m}{RT}+\frac{a}{V_mRT}=1$$

$$\boxed{Z=1-\frac{a}{V_mRT}} \quad \Rightarrow \quad Z<1$$

As pressure increases, V_m decreases and hence Z decreases.

(ii) At high pressure and constant temperature

At high pressure, V_m will be low. So 'b' can't be neglected in comparision to V_m but $\frac{a}{V_m^2}$ can be neglected as compared to much higher values of P. Thus equation (1) reduces to

$$P(V_m-b)=RT$$

$$PV_m-Pb=RT$$

$$\frac{PV_m}{RT}=\frac{Pb}{RT}+1$$

$$\boxed{Z=1+\frac{bP}{RT}} \Rightarrow (Z>1)$$

On increasing the pressure, Z increases.

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(iii) At low pressure and very high temperature.

V_m will be very large, hence 'b' can be neglected and $\frac{a}{V_m^2}$ can also be neglected as V_m is very large.

$PV_m = RT$ (ideal gas condition)

(iv) For H_2 or He $a \simeq 0$ because molecules are smaller in size or vander Waal's forces will be very weak.

$P(V_m - b) = RT$

So $\mathbf{Z = 1 + \frac{Pb}{RT}}$

This explains type I plot.

1.5 LIMITATIONS OF VANDER WAAL'S GAS EQUATION :

(i) Vander waal considered only attractive forces in gas molecules, but in reality, the molecules experience attractive as well as repulsive forces. The positive deviation ($Z > 1$) was explained due to finite size of particles ('b' is not negligible relative to V_m)

(ii) The constants 'a' and 'b' are considered as characteristic constants for a gas, independent from temperature and pressure. But in reality, they should change with conditions.

(iii) At critical point (discussed later in this sheet, 3.2), Z becomes 3/8 for all gases obeying Vander waal's gas equation. But in reality, different gases have different values of Z at critical point and none of them have the value 3/8.

1.6 SOME IMPORTANT INFORMATIONS :

(i) There is no any general equation relating P, V and T of gases, obeyed by all the gas, in all conditions.

(ii) Gases obeying PV = nRT in all the conditions are called **ideal** or **perfect** gases and none of the gas in nature is ideal gas.

(iii) Gases obeying $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ in all the conditions are called **Vander waal** gases and none of the gas in nature is Vander waal gas.

(iv) **Virial form of gas equation :**

$$PV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + ...\right)$$

Where B = second virial coefficient

C = third virial coefficient

Ex.1. Calculate the pressure exerted by 5 mole of CO_2 in one litre vessel at 47°C using van der waal's equation. Also report the pressure of gas if it behaves ideal in nature.

Given that a = 3.592 atm lt^2 mol^{-2}, b = 0.0427 L/mol. Also, if the volume occupied by CO_2 molecules is negligible, then calculate the pressure exerted by one mole of CO_2 gas at 273 K.

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Sol. Vander waal's equation

$$\left[p+\frac{n^2a}{V^2}\right][V - nb] = nRT$$

$n_{CO_2} = 5$, V = 1 litre, T = 320 K, a = 3.592, b = 0.0427

$$\therefore \quad \left[P+25\times\frac{3.592}{1}\right][1 - 5 \times 0.0427] = 5 \times 0.0821 \times 320$$

$$\therefore \quad P = 77.218 \text{ atm}$$

For ideal behaviour of gas, PV = nRT

$$\therefore \quad P \times 1 = 5 \times 0.0821 \times 320$$

$$\therefore \quad P = 131.36 \text{ atm}$$

For one mole $\left[P+\frac{a}{V^2}\right][V - b] = RT$

$$\therefore \quad P = \frac{RT}{V} - \frac{a}{V^2}$$

$$\therefore \quad P = \frac{0.0821\times273}{22.4} - \frac{3.592}{(22.4)^2}$$

$$\therefore \quad P = 0.9922 \text{ atm}$$

The volume occupied by 1 mole at 273 K is 22.4 litre if b is negligible.

Ex.2 One mole of CCl_4 vapours at 77°C occupies a volume of 35.0 L. If vander waal's constants are a = 20.39 L^2 atm mol^{-2} and b = 0.1383 L mol^{-1}, calculate compressibility factor Z under, (a) low pressure region. (b) high pressure region.

Sol. (a) Under low pressure region, V is high

$$\therefore \quad (V - b) \approx V$$

$$\left(P+\frac{a}{V^2}\right)V = RT$$

$$PV + \frac{a}{V} = RT$$

$$\frac{PV}{RT}+\frac{a}{RTV} = 1$$

$$Z = \frac{PV}{RT} = \left(1-\frac{a}{RTV}\right) = 1 - \frac{20.39}{0.0821\times350\times35} = 0.98$$

(b) Under high pressure region, P is high,

$$\left(P+\frac{a}{V^2}\right) \approx P$$

$$\therefore \quad P (V - b) = RT$$

$$PV - Pb = RT$$

$$\frac{PV}{RT}-\frac{Pb}{RT} = 1$$

$$\therefore \quad Z = \frac{PV}{RT} = 1 + \frac{Pb}{RT} \qquad \left(\because \frac{PV}{RT} = 1, \text{ or } \frac{P}{RT} = \frac{1}{V}\right)$$

$$Z = 1 + \frac{b}{V} = 1 + \frac{0.1383}{35} = 1 + 0.004 = 1.004$$

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***Ex.3** One way of writing the equation of state for real gases is PV = RT $\left[1+\frac{B}{V}+\ldots\right]$ where B is a constant and V is the molar volume of gas. Derive an expression for B in terms of van der Waal's constants a and b.*

Sol. According to van der Waal's equation

$$\left[P+\frac{a}{V^2}\right][V-b] = RT \text{ or } \quad P=\frac{RT}{(V-b)}-\frac{a}{V^2}$$

Multiply by V, then

$$PV = \frac{RTV}{(V-b)}-\frac{a}{V} \quad \text{or} \quad PV = RT\left[\frac{V}{V-b}-\frac{a}{VRT}\right]$$

or $$PV = RT\left[\left(1-\frac{b}{V}\right)^{-1}-\frac{a}{VRT}\right]$$

$\because$ $$\left[1-\frac{b}{V}\right]^{-1} = 1+\frac{b}{V}+\left(\frac{b}{V}\right)^2+\ldots.$$

$\therefore$ $$PV = RT\left[1+\frac{b}{V}+\ldots-\frac{a}{VRT}\right]$$

$$PV = RT\left[1+\left(b-\frac{a}{RT}\right)\cdot\frac{1}{V}+\ldots\right]$$

Comparing with the given equation :

$\therefore$ $$B = b-\frac{a}{RT}$$

2. BOYLE TEMPERATURE :

(i) It is the temperature at which a real gas behave ideally in a wide range of pressure, in low pressure region.

(ii) (a) Tamperature < Boyle temperature

Z < 1, low pressure range

Z > 1, high pressure range

(b) Temperature = Boyle temperature

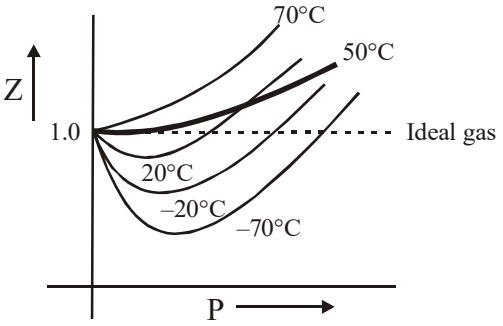
Z = 1, low pressure range

Z > 1, high pressure range

(c) At temperature > Boyle temperature

Z > 1, at all pressure

(d) $T \to \infty$, $Z \to 1$



On increasing the temperature, the thermal energy increases and simultaneously the attractive forces decreases. Hence a temperature comes at which the thermal energy become too high that it balances the effect of attraction and gas molecules becomes independent.

If at Boyle temperature, pressure is increased, molecules come more closer. Due to repulsive force, Z becomes greater that 1.

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2.1 CALCULATION OF T_B :

(i) $PV_m = RT = \left(1 + \frac{B}{V_m} + \underbrace{\frac{C}{V_m^2} + ...}_{\text{negligible}}\right)$ at low pressure.

At T = T_B, the second virial coefficient should be 0

$B = 0$

or $b - \frac{a}{RT_B} = 0$

$\therefore$ $\boxed{T_B = \frac{a}{Rb}}$

(ii) **Calculus method :**

At Boyle temperature, $\left(\frac{\partial Z}{\partial P}\right)_T = 0$ at low pressure.

Ex.4 Derive the expresion for compresibility factor of a vanderwaal gas at Boyle temperature.

Solution : $Z = \frac{V_m}{V_m - b} - \frac{a}{V_m - RT}$

At Boyle temperature,

$Z = \frac{V_m}{V_m - b} - \frac{a}{V_m R \times \frac{a}{Rb}}$

$Z = \frac{V_m}{V_m - b} - \frac{b}{V_m}$

$Z = \frac{V_m^2 - V_m b + b^2}{V_m (V_m - b)}$

$Z = 1 + \frac{b^2}{V_m (V_m - b)}$

Ex.5 Calculate the volume occupied by 2 moles of a vanderwall gas at 5 atm 800 K.

Given : a = 4.0 atm $\ell^2 mol^{-2}$, b = 0.0625 $\ell\, mol^{-1}$, R = 0.08 ℓ -atm/K-mol

Solution : $T_B = \frac{a}{Rb} = 800K$

Gas behave ideally at given condition.

PV = nRT

$5 \times V = 2 \times 0.08 \times 800$

V = 25.6 litre

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3. LIQUEFACTION OF GASES AND CRITICAL CONSTANTS :

The phenomenon of converting a gas into liquid is known as liquefaction. The liquefaction of a gas takes place when the intermolecular forces of attraction become so high that they exist in liquid state. A gas can be liquefied by :

(a) **Increasing pressure :** An increase in pressure results decrease in intermolecular distance.

(b) **Decreasing temperature :** A decrease in temperature results decrease in kinetic energy of molecules.

Note : Due to absence of intermolecular forces, ideal gases can never be liquified.

3.1 Andrews Isotherms :

The essential conditions for liquefaction of gases were discovered by Andrews (1869) as a result of his study of P–V–T relationship for CO_2.The types of isotherms are shown in figure.

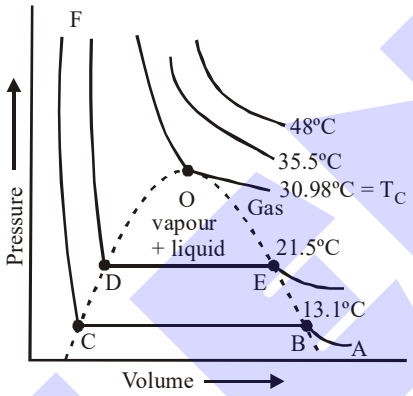


Fig : Isotherms for carbon dioxide

Observations from figure :

(a) **At low temperatures :**

For the curve ABCF, as the pressure increases, volume of the gas decreases (curve A to B). At point B, at constant pressure, liquefaction commences and the volume decreases rapidly (because gas is converted to liquid with higher density). At point C, liquefaction is complete. The line CF represents the variation of V with P of the liquid state. The steepness of the line CF indicates that the liquid cannot be easily compressed. Thus AB, represent gaseous state, BC represent liquid and vapour in equilibrium and CF represent liquid state.

The pressure corresponding to the line BC is vapour pressure of the liquid at that temperature.

(b) **At lower temperatures** :

Similar type of curve as in case (A) is obtained but the width of the horizontal portion is reduced. The pressure corresponding to this portion is higher than at lower temperatures.

(c) **At high temperatures** :

(say 48°C), the isotherms are like those of ideal gas. Gas does not liquify, even at very high pressure.

(d) **At temperature (30.98°C)** : The horizontal portion is reduced to a point.

The isotherm at T_C is called **critical isotherm**.

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3.2 Critical parameters or critical constants :

(i) **Critical temperature (T_C)** :

The temperature above which a system can never be liquefied by the application of pressure alone i.e. the temperature above which the liquid cannot exit is called the critical temperature T_C.

(ii) Critical pressure (P_C) :

The minimum pressure required to liquefy the gas at critical temperature (T_C) is called the critical pressure, P_C.

(iii) Critical volume (V_C) :

The volume occupied by one mole of the system at critical temperature, (T_C) and critical pressure, (P_C) is called the critical volume (V_C) of the gas.

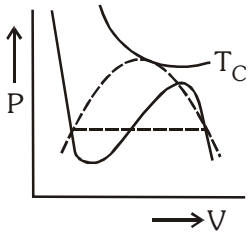
T_C, P_C and V_C are collectively called **critical parameters** or **critical constants of the gas.**

3.3 Determination of value of P_C, V_C and T_C :

The Vander waal's equation is

$$\left(P+\frac{a}{V_m^2}\right)(V_m - b) = RT$$

or $$V_m^3 - \left(b+\frac{RT}{P}\right)V_m^2 + \frac{a}{P}V_m - \frac{ab}{P} = 0 \quad \ldots. (1)$$



This equation has three roots in V_m for given values of a, b, P and T. It is found that either all the three roots are real or one is real and the other two are imaginary.

At temperature lower than T_C , the isotherm exhibits a maximum and a minimum for certain values of pressures, the equation gives three real roots of volume e.g., V_1, V_2 and V_3 at pressure P_1. On increasing the temperature, the three roots become closer to each other and ultimately at critical temperature, they become identical. Thus, the cubic equation V_m can be written as

$$(V_m - V')(V_m - V'')(V_m - V''') = 0$$

At the critical point $V' = V'' = V''' = V_C$, and hence the equation becomes,

$$(V_m - V_C)^3 = 0$$

or $$V_m^3 - V_C^3 - 3V_C V_m^2 + 3V_C^2 V_m = 0 \quad \ldots..(2)$$

By comparing the coefficients in eq.(1) and eq(2)

$$3V_C = b + \frac{RT_C}{P_C}, \quad 3V_C^2 = \frac{a}{P_C}, \quad V_C^3 = \frac{ab}{P_C}$$

By solving, $\mathbf{V_C = 3b}$, $\mathbf{P_C = \frac{a}{27b^2}}$ and $\mathbf{T_C = \frac{8a}{27Rb}}$

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3.4 The value of critical compressibility factor in terms of vander waal's constants is given by

$$Z = \frac{P_C V_C}{RT_C} = \frac{\frac{a}{27b^2} \times 3b}{R \times \frac{8b}{27Rb}} = \frac{3}{8} = 0.375$$

If we compare the value of $\frac{P_C V_C}{RT_C} = 0.375$, with the experimental values, it has been found that the agreement is very poor.

Critical constants of gases

Substance	T_c / K	P_c / bar	V_c / dm³ mol⁻¹
H_2	33.2	12.97	0.0650
He	5.3	2.29	0.0577
N_2	126.0	33.9	0.0900
O_2	154.3	50.4	0.0744
CO_2	304.10	73.9	0.0956
H_2O	647.1	220.6	0.0450
NH_3	405.5	113.0	0.0723

Ex.6 The critical temperature and pressure of CO_2 gas are 304.2 K and 72.9 atm respectively. What is the radius of CO_2 molecule assuming it to behave as vander Waal's gas ?

Sol. $T_C = 304.2$ K $P_C = 72.9$ atm

$$T_C = \frac{8a}{27Rb} \qquad P_C = \frac{a}{27b^2}$$

$$\therefore \quad \frac{T_C}{P_C} = \frac{\frac{8a}{27Rb}}{\frac{a}{27b^2}} = \frac{8a}{27Rb} \times \frac{27b^2}{a} = \frac{8b}{R} \quad \text{or} \quad b = \frac{RT_C}{8P_C} = \frac{1}{8} \times \frac{0.082 \times 304.2}{72.9} = 0.04277 \text{ lit mol}^{-1}$$

$$b = 4\, N_A \times \frac{4}{3}\pi r^3 = 42.77 \text{ cm}^3$$

or $r = (4.24)^{1/3} \times 10^{-8}$ cm $= 1.62 \times 10^{-8}$ cm

$\therefore$ radius of CO_2 molecule = 1.62 Å

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SELF STUDY

4. THE LIQUID STATE

Liquid state is intermediate between gaseous and solid states. The liquids posses fluidity like gases but incompressibility like solids.

The behaviour of liquids explained above gives some characteristic properties to the liquids such as definite shape, incompressibilty, diffusion, fluidity (or viscosity), evaporation (or vapour pressure), surface tension, etc.

The following general characteristics are exhibited by liquids :

(i) Shape :

Liquids have no shape of their own but assume the shape of the container in which they are kept. No doubt, liquids are mobile but they do not expand like gases as to fill up all the space offered to them but remain confined to the lower part of the container.

(ii) Volume :

Liquids have definite volume as the molecules of a liquid are closely packed and the cohesive forces are strong.

(iii) Density :

As the molecules in liquids are closely packed, the densities of liquids are much higher than in gaseous state. For example, density of water at 100° C and 1 atmospheric pressure is 0.958 g mL^{-1} while that of water vapour under similar conditions as calculated from ideal gas law $\left(d = \frac{MP}{RT}\right)$ is 0.000588 g mL^{-1}.

(iv) Compressibility :

The molecules in a liquid are held in such close contact by their mutual attractive forces (cohesive forces) that the volume of any liquid decreases very little on increasing pressure. Thus, liquids are relatively incompressible compared to gases.

(v) Diffusion :

When two miscible liquids are put together, there is slow mixing as the molecules of one liquid move into the other liquid. As the space available for movement of molecules in liquids is much less and their velocities are small. Liquids diffuse slowly in comparison to gases.

(vi) Evaporation :

The process of changes of liquid into vapour state on standing is termed **evaporation**. Evaporation may be explained in terms of motion of molecules. At any given temperature all the motion of molecules do not possess the same kinetic energy (average kinetic energy is, however same). Some molecules move slowly, some at intermediate rates and some move very fast. A rapidly moving molecule near the surface of the liquid may possess sufficient kinetic energy to overcome the attraction of its neighbours and escape. Evaporation depends on the following factor :

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(a) Nature of the liquid : The evaporation depends on the strength of intermolecular forces (cohesive forces). The liquids having low intermolecular forces evaporate faster in comparison to the liquids having high intermolecular forces. For example, ether evaporates more quickly than alcohol, and alcohol evaporates more quickly than water, as the intermolecular forces in these liquids are in the order :

Ether < Alcohol < Water

(b) Surface area : Evaporation is a surface phenomenon. Larger the surface area, greater is the oportunity of the molecules to escape. Thus, rate of evaporation increases with increase of surface area.

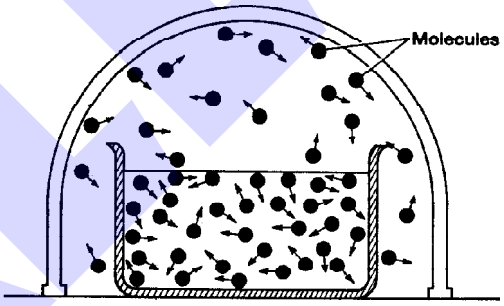
(c) Temperature : Rate of evaporation increases with the increase of temperature as the kinetic energy of the molecules increases with the rise of temperature.

(d) Flow of air current over the surface : Flow of air helps the molecules to go away from the surface of liquid and, therefore, increases the evaporation of liquid in open vessel.

(vii) Latent heat of vaporisation :

The quantity of heat required to vaporise unit mass of a given liquid at constant temperature is known as latent heat of vaporisation. The latent heat of vaporisation depends on the strength of the intermolecular forces with in the liquid. The value of latent heat of vaporisation decreases with increase in temperature and it becomes zero at the critical temperature. When the vapour is condensed into a liquid, heat is evolved. This is called **heat of condensation**. It is numerically equal to the heat of vaporisation at the same temperature.

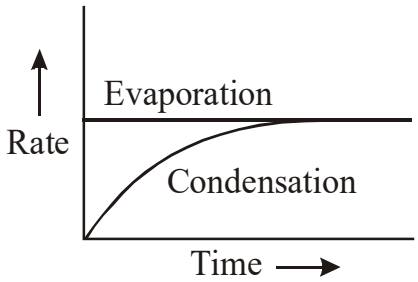
(viii) Vapour pressure :



When the space above the liquids is closed, the molecules cannot escape into open but strike the walls of the container, rebound and may strike the surface of the liquid, where they may be trapped. The return of the molecules from the vapour state of the liquid state is known as **condensation**. As evaporation proceeds, the number of molecules in the vapour state increases and, in turn, the rate of condensation increases.

The rate of condensation soon becomes equal to the rate of the evaporation, i.e., the vapour in the closed container is in equilibrium with the liquid.

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At equilibrium the concentration of molecules in the vapour phase remains unchanged. The pressure exerted by the vapour in equilibrium with liquid, at a given temperature, is called the **vapour pressure**.

Since the rate of evaporation increases and rate of condensation decreases with increasing temperature, vapour pressure of liquids always increases as temperature increases. At any given temperature, the vapour pressures of different liquids are different because their cohesive forces are different. Easily vaporisable liquids are called **volatile liquids** and they have relatively high vapour pressure. Vapour pressure values (in mm of Hg) for water, alcohol and ether at different temperatures are given in the following table :

Substance	**Temperatures**				
	0° C	20° C	40° C	80° C	100° C
Water	4.6	17.5	55.0	355.5	760.3
Ethyl alcohol	12.2	43.9	812.6	1693.3	
Diethyl ether	185.3	442.2	921.1	2993.6	4859.4

The vapour pressure of a given liquid at two differnt temperatures can be compared with the help of **Clausius-Clapeyron equation.**

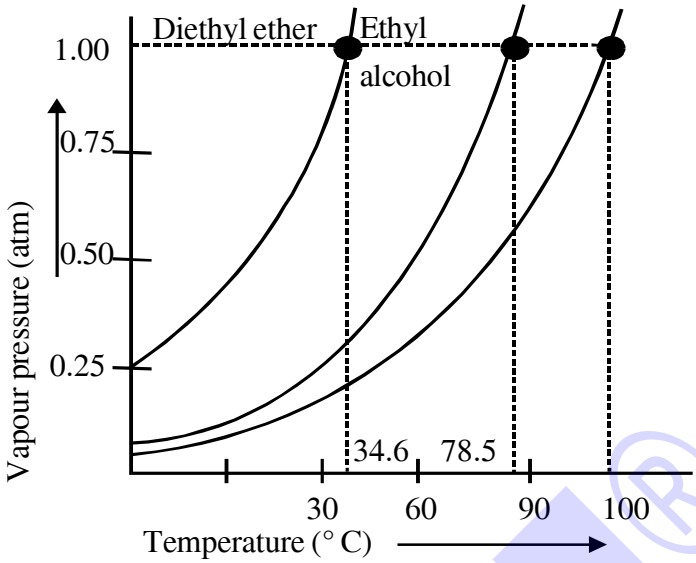
$$\ell n\frac{P_2}{P_1}=\frac{\Delta H}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

Where ΔH is the latent heat of vaporisation and R is the molar gas constant.

(ix) Boiling point :

The temperature at which the vapour pressure of the liquid becomes equal to the atmospheric pressure is called the **boiling point** of the liquid. When a liquid is heated under a given applied pressure, bubbles of vapour begin to form below the surface of the liquid. They rise to the surface and burst releasing the vapour into the air. This process is called **boiling**. The normal boiling point is the temperature at which the vapour pressure of a liquid is equal to one atmospheric pressure (760 mm of Hg). Figure shows that normal boiling points of di-ethyl ether, ethyl alcohol and water are 34.6° C, 78.5° C and 100° C respectively.

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The temperature of the liquid remains constant until all the liquid has been vaporised. Heat must be added to the boiling liquid to maintian the temperature because in the boiling process, the high energy molecules are lost by the liquid.

The boiling point of a liquid changes with the change in external pressure. A liquid may boil at temperatures higher than normal under external pressures greater than one atmoshpere. Conversely, the boiling point of a liquid may be lowered than normal below one atmosphere. Thus, at high altitudes where the atmospheric pressure is less than 760 mm, water boils at temperatures below its normal boiling water.

Boiling and evaporation are similar processes (conversion of liquid into vapour) but differ in following respects :

(a) Evaporation takes place spontaneously at all temperatures but boiling occurs at a particular temperature at which the vapour prssure is equal to the atmospheric pressure.

(b) Evaporation is surface phenomenon. It occurs only at the surface of the liquid whereas boiling involves formation of bubbles below the surface of the liquid.

Note. : Boiling does not occur when liquid is heated in a closed vessel.

(x) Freezing point :

When a liquid is cooled, the kinetic energy of the molecules goes on decreasing. A stage comes when the intermolecular forces become stronger and the rotational motion is siezed. At this stage, the formation of solid begins and the liquid is seen to freeze out. The point (temperature) at which the vapour pressure of solid and liquid forms of a substance become equal is termed as **freezing point.**

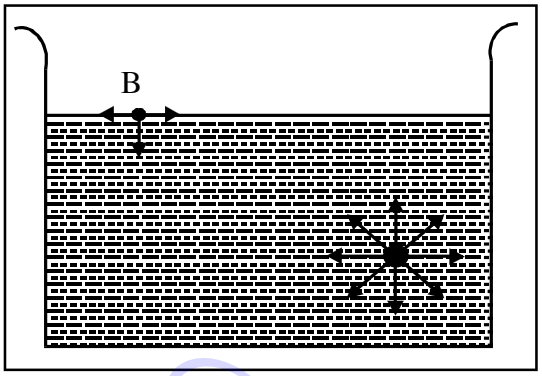
Normal freezing point of a liquid is the temperature at which is liquid and solid forms are in equilibrium with each other under a pressure of one atmosphere. The freezing point of a liquid is the same as the melting point of its solid. The amount of heat that must be removed to freeze a unit mass of the liquid at the freezing point, which is called the **latent heat of fusion.**

The freezing point of a liquid is affected by the change of external pressure. With increased external pressure, the freezing point of some liquids rises while of others falls. But the effect of pressures is very small because solid as well as liquid are almost incompressible.

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(xi) Surface tension :

It is the property of liquids caused by the intermolecular attractive forces. A molecule within the bulk of the liquid is attracted equally in all the directions by the neighbouring molecules. The resultant force on any one molecule in the centre of the liquid is, therefore, zero. However, the molecules on the surface of the liquid are attracted only inward and sideways. This unbalanced molecular attraction pulls some of the molecules into the bulk of the liquid, i.e., are pulled inward and the surface area is minimized.



Surface tension is a measure of this inward force on the surface of the liquid. It acts downwards perpendicular to the plane of the surface. The unit of surface tension is dyne cm^{-1}. Surface tension is, defined **as the force acting on the surface at right angles to any line of unit length.**

As the intermolecular forces of attraction decreases with the rise of temperature, the surface tension of a liquid, thus, decreases with increase in temperature. Similarly, addition of chemicals to a liquid may reduce its surface tension. For example, addition of chemicals like soaps, detergents, alcohol, camphor, etc., lowers the surface tension of water.

Many common phenomenon can be explained with the help of surface tension. Some are described here :

(a) Small droplets are spherical in shape :

The surface tension acting on the surface of the liquid tries to minimise the surface area of a given mass of a liquid. It is known that for a given volume, a sphere has the minimum surface area. On account of this, drops of liquids acquire a spherical shape.

(b) Insects can walk on the surface of water :

Many insects can walk on the surface of water without drowing. This is due to the existence of surface tension. The surface tension makes the water surface to behave like an elastic membrane and prevents the insects from drowning.

(c) Cleaing action of soap and detergents :

Soap and detergent solutions due to their lower surface tensions penetrate into the fibre and surround the greasy substances and wash them away.

(d) Capillary action :

The tendency of a liquid to rise into narrow pores and tiny openings is called capillary action. The liquids rise in the capillary tubes due to the surface tension.

(e) Surface Energy :

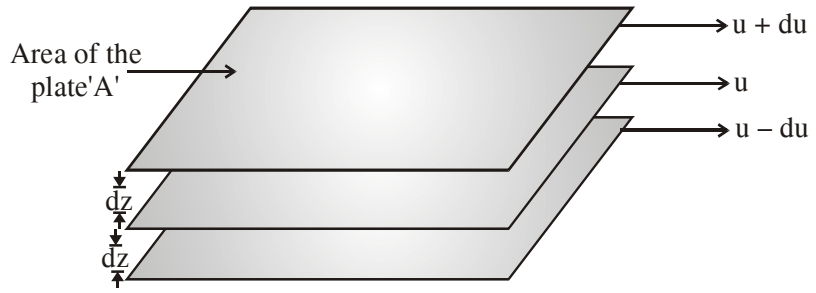
The work required to be done to increase or extend surface area by unit area is called surface energy. The units of surface energy is joule per sq. metre, i.e. J m^{-2} in S.I. system.

(xii) Viscosity :

Viscosity is a measure of resistance to flow which arises due to the internal friction between layers of fluid as they slip past one another while liquid flows. Strong intermolecular forces between molecules hold them together and resist movement of layers past one another.

When a liquid flows over a fixed surface, the layer of molecules in the immediate contact of surface is stationary. The velocity of upper layers increase as the distance of layers from the fixed layer increases. This type of flow in which there is a regular gradation of velocity in passing from one layer to the next is called **laminar flow**. If we choose any layer in the flowing liquid , the layer above it accelerates its flow and the layer below this retards its flow.

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Gradation of velocity in the laminar flow

If the velocity of the layer at a distance dz is changed by a value du then velocity gradient is given by the amount $\frac{du}{dz}$. A force is required to maintain the flow of layers. This force is proportional to the area of contact of layers and velocity gradient i.e.

$F \alpha A$ (A is the area of contact)

$F \alpha A \frac{du}{dz}$ (where, $\frac{du}{dz}$ is velocity gradient; the change in velocity with distance)

$F \alpha A. \frac{du}{dz} \Rightarrow F = \eta A \frac{du}{dz}$

'η' is proportionality constant and is called **coefficient of viscosity**. Viscosity coefficient is the force when velocity gradient is unity and the area of contact is unit area. SI unit of viscosity coefficient is 1 newton second per square metre (N s m^{-2}) = pascal second (Pa s = 1kg $m^{-1}s^{-1}$).

Greater the viscosity, the more slowly the liquid flows. Hydrogen bonding and van der Waals forces are strong enough to cause high viscosity. Glass is an extremely viscous liquid. It is so viscous that many of its properties resemble solids. However, property of flow of glass can be experienced by measuring the thickness of windowpanes of old buildings. These become thicker at the bottom than at the top.

Viscosity of liquids decreases as the temperature rises because at high temperature molecules have high kinetic energy and can overcome the intermolecular forces to slip past one another between the layers.

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SOME MISCELLANEOUS SOLVED EXAMPLES

1. Calculate the temperature of gas if it obeys vander waal's equation from the following data. A flask of 25 litre contains 10 moles of a gas under 50 atm. Given a = 5.46 atm litre^{-2} and b = 0.31 litre mol^{-1}.

Sol. Given , P = 50 atm, V = 25 litre, n = 10

$a = 5.46$ atm litre^{2}mol^{-2}, $b = 0.031$ litre mol^{-1}

Now vander waal's equation for n mole of gas.

$$\left[P + \frac{n^2a}{V^2}\right][V - nb] = nRT$$

$$\left[50 + \frac{100 \times 5.46}{625}\right][25 - 10 \times 0.031] = 10 \times 0.0821 \times T$$

$\therefore$ T = 1529.93 K = 1256.93ºC

2. The molar volume of He at 10.135 MPa and 273 K is 0.011075 of its molar volume of 101.325 kPa at 273K. Calculate the radius of helium atom. The gas is assumed to show real gas nature. Neglect the value of a for He.

Sol. For real gas : $\left[P + \frac{a}{V^2}\right][V - b] = RT$

$P[V - b] = RT$ (neglecting a)

$$\therefore \frac{10.1325 \times 10^6}{101325}[V_1 - b] = 0.0821 \times 273$$

or $100[V_1 - b] = 0.0821 \times 273 = 22.41$ .....(1)

$$\frac{101.325 \times 10^3}{101325}[V_2 - b] = 0.0821 \times 273$$

or $[V_2 - b] = 22.41$ .....(2)

by eq. (1) $V_1 - = 0.2241 + b$ .....(3)

by eq. (2) $V_2 - = 22.41 + b$ .....(4)

By eqs. (3) and (4), $\frac{V_1}{V_2} = \frac{0.2241 + b}{22.41 + b}$

$$\frac{0.011075V_2}{V_2} = \frac{0.2241 + b}{22.41 + b}$$

($V_1 = 0.011075V_2$ is given)

$\therefore$ b = 0.024 litre mol^{-1} = 24 cm^{3}mol^{-1}

$$\therefore b = 4N \times v = 4 \times 6.023 \times 10^{23} \times \frac{4}{3}\pi r^3$$

$$\text{or } 24 = 4 \times 6.023 \times 10^{23} \times \frac{4}{3} \times \frac{22}{7} \times r^3$$

$\therefore$ **r = 1.33 × 10^{-8} cm**

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3. Calculate vander waal's constant a and b if critical temperature and critical pressure are 30°C and 72 atm respectively.

Sol. $T_C = \frac{8a}{27Rb}$ and $P_C = \frac{a}{27b^2}$

$\therefore \frac{T_C}{P_C} = \frac{8b}{R}$ or $b = \frac{RT_C}{8P_C}$

$\therefore b = \frac{0.0821 \times 303}{8 \times 72} = 0.043$ litre mol^{-1}

$\alpha = 27P_C \times b^2 = 27 \times 72 \times (0.043)^2$

$= 3.59$ litre^{2}atm mol^{-2}

4. Using vander waal's equation, calculate the constant, 'a' when two mole of a gas confined in a four litre temperature of 300K. The value of 'b' is 0.05 litre mol^{-1}.

Sol. Vander waal's equation for n mole of gas is

$\left[P + \frac{n^2a}{V^2}\right]$ [V-nb]=nRT

Given V = 4 litre ; P = 11.0 atm, T = 300 K

b = 0.05 litre mol^{-1} ; n =2

Thus, $\left[11 + \frac{2^2a}{4^2}\right]$ [4 – 2 × 0.05] = 2 × 0.082 × 300.

$\therefore$ a = 6.46 atm litre^2 mol^2

5. One way of writing the equation for state for real gases is, **[JEE 1997]**

$P\bar{V} = RT\left[1 + \frac{B}{\bar{V}} + ...\right]$ ***where B is a constant.***

Derive an approximate expression for 'B' in terms of van der Waals' constants 'a' & 'b'.

Ans. $B = \left(b - \frac{a}{RT}\right)$

Sol. $\left[P + \frac{a}{Vm^2}\right]$ [Vm – b]

$P = \frac{RT}{Vm - b} - \frac{a}{Vm^2}$

$\frac{PVm}{RT} = \frac{Vm}{Vm - b} - \frac{a}{VmRT}$

$Z = \left(1 - \frac{b}{Vm}\right)^{-1} - \frac{a}{VmRT}$

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$$= 1 + \frac{b}{Vm} + \frac{b^2}{Vm^2} + ... - \frac{a}{VmRT}$$

$$= 1 + \left[b - \frac{a}{RT}\right] \frac{1}{Vm} + \frac{b^2}{Vm^2} .....+ \quad ... (1)$$

Also, $P\bar{V} = RT\left[1 + \frac{B}{V} + ...\right]$

$$\Rightarrow \frac{P\bar{V}}{RT} = \left[1 + \frac{B}{V} + ...\right] \quad .... (2)$$

$\Rightarrow$ Comparing equation (1) & (2), we get

$$B = b - \frac{a}{RT}$$

6. Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atm at a temperature of 300 K. The value of "b" is 0.05 litre mol^{-1}. **[JEE 1998]**

Ans. 6.52 atm L^2 mol^{-2}

Sol. $\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$

$$\left[11 + \frac{2^2 \times a}{4 \times 4}\right][4 - 2 \times 0.05] = 2 \times 0.0821 \times 300$$

$\Rightarrow a = 6.52$ atm L^2 mol^{-2}

7. A gas will approach ideal behaviour at : **[JEE 1999]**

(A) low temperature and low pressure

(B) low temperature and high pressure

(C) low pressure and high temperature

(D) high temperature and high pressure .

Ans. (C)

Sol. A gas approches ideal behavior at High T & low P

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8. ***The compressibility of a gas is less than unity at STP. Therefore ,*** **[JEE 2000]**

(A) $V_m > 22.4\ L$ (B) $V_m < 22.4\ L$

(C) $V_m = 22.4\ L$ (D) $V_m = 44.8\ L$

Ans. (B)

Sol. $Z = \frac{(Vm)real}{(Vm)ideal}; Z < 1$

$\Rightarrow (Vm)_{real} < (Vm)_{ideal}$

$\therefore (Vm)_{real} < 22.4\ L$

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